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Professor Roman Słowiński
Co-ordinating Editor-in-Chief
European Journal of Operational Research
Institute of Computing Science, Poznań University of Technology
60-965 Poznań, Poland

Dear Professor Słowiński,

Please find enclosed our manuscript, **"OR-Select: Reproducible Best-of-$N$ Answer Selection over LLM Ensembles, with an Exact Kemeny Reference and a Ground-Truth Audit,"** submitted for consideration as a Research Article in the *European Journal of Operational Research*.

Sampling several answers from a large language model and choosing one is now standard practice, but the usual choosing rules are either majority vote, which is undefined when answers do not repeat, or an LLM judge whose verdicts are not reproducible. We formulate selection as an integer program over a judge-derived pairwise comparison matrix and a deterministic self-consistency term, solved exactly with CP-SAT, so that the winner is exactly reproducible and every stored run can be replayed offline. The Copeland-style aggregate we select on inherits the 5-approximation guarantee of Coppersmith, Fleischer, and Rudra against the NP-hard Kemeny-optimal ranking, and we compute the exact Kemeny–Young winner on every problem as a reference.

The manuscript's contribution is an audit rather than a claim of superiority. Against self-consistency and greedy sampling on six public benchmarks (GSM8K, MATH Level 5, AIME 2026, GPQA Diamond, AIME 2025, MMLU-Pro; 532 problems), using paired exact tests and a floor/oracle recovery metric, we find no accuracy benefit of judge-based selection with a single generator and a significant harm, pooled over the four heterogeneous tracks, with heterogeneous generators, and we trace the latter to a length-scale mismatch in the default objective and a generator-specific preference of the judge, using offline replays that cost no API calls. We believe this is of interest to *EJOR*'s readership at the intersection of rank-aggregation theory, integer programming, and the evaluation of language-model systems, and we report the study's limitations directly: a single free-tier judge model, small candidate sets, post hoc diagnostic analyses, and a heterogeneous pool that mixes models because one provider model was withdrawn during the study.

Code, stored run files and generated tables are archived on Zenodo, DOI 10.5281/zenodo.24112045.

All authors have approved the manuscript and agree to its submission to *European Journal of*

*Operational Research*. We have no conflicts of interest to disclose. Thank you for your consideration.

Sincerely,

Aris Dressino